GPSTEC317

AWS re:INVENT

From Leaves to Lawns

AWS Greengrass at the Edge and Beyond

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Alex González, Engineering Supervisor, Digi International

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Agenda

AWS Greengrass overview and case studies

AWS Greengrass runtime components

Developing AWS Greengrass apps for scale

Demonstration

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Capabilities and Case Studies

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AWS IoT

Things
Sense & Act

Cloud
Storage & Compute

Intelligence
Insights & Logic → Action

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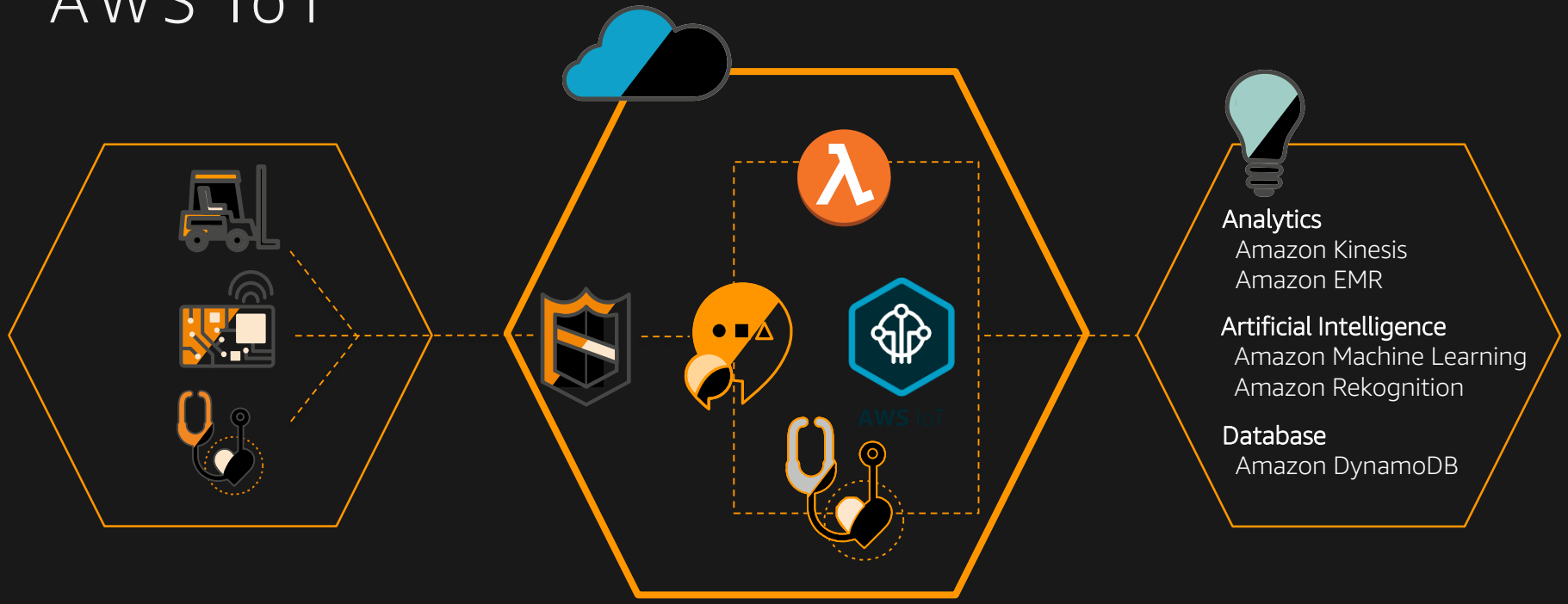
AWS IoT



Things

Cloud

Intelligence

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Value of processing data at the source

Law of physics

Law of economics

Law of the land

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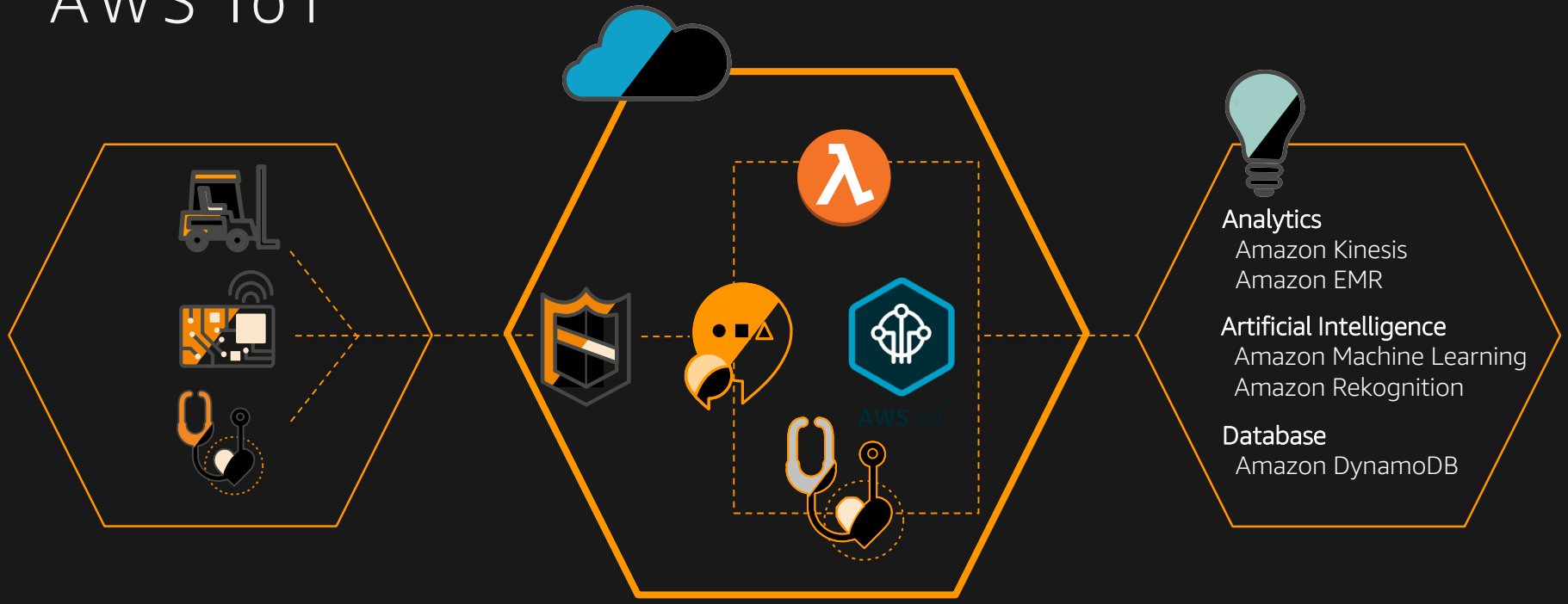
AWS IoT



Things

Cloud

Intelligence

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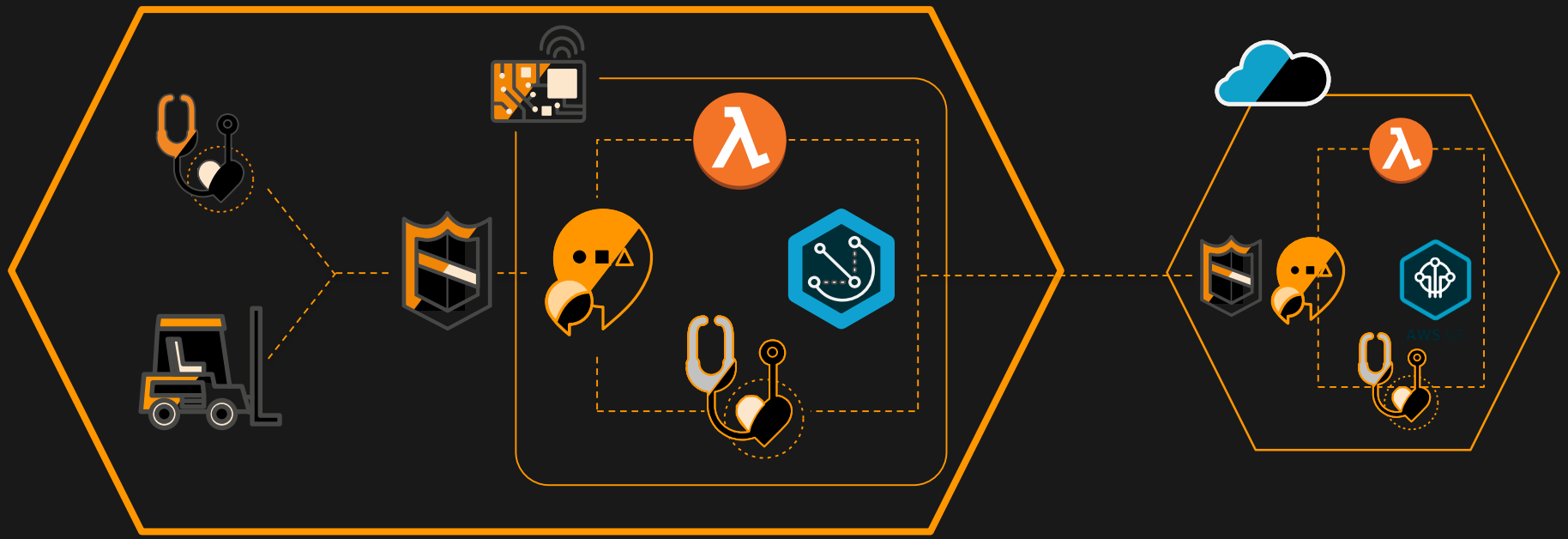
AWS Greengrass



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Key customer edge challenges

Responding quickly to local events

Operating offline

Simplifying device programming

Reducing costs of IoT applications

World class security

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RioTinto

Problem

Rio Tinto has connectivity challenges at some of the mine locations where large, expensive machinery is in play. Rio was looking for a way to still leverage the cloud to predict failures and learn from their environment.

Solution

Rio is using AWS Greengrass to calculate road roughness from sensors on haul trucks and create an online heat map of the rough roads. Maintenance crews will use this information to effect road repairs and reduce premature damage of their machinery.

Impact

Unlike current on-premise programs for monitoring the machinery, AWS Greengrass allows for real-time alerts and machine-to-machine communication while leveraging Machine Learning models in the cloud when connectivity is available.

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Problem

Nokia has seen a need in industrial IoT to analyze video streams at the edge and send the data to remote centers only when anomalies are detected.

Solution

Deploying AWS Greengrass on Nokia Multi-access Edge Computing platform and combining it with Nokia private mobile network solutions. This joint solution will make it possible for the oil industry to pair real time drilling data with production data of nearby wells.

Impact

Due to the cost of bandwidth being expensive, this allows Nokia to optimize the data that is sent to other wells and to the cloud based on rules and alerts set up on the locally-processed data.

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Problem

Pentair provides beer and water filtration systems to large industrial brewing customers like Heineken. Most of their industrial customers are located in remote geographies with unreliable internet connections. They also have customers who do not want to open up firewalls port for internet connectivity.

Solution

They want to phase out or integrate their current PLCs with AWS Greengrass clusters to make real-time decisions on premise and eventually streaming to the cloud for further analysis.

Impact

Pentair can take this use case and replicate it across their various workloads in commercial, residential and industrial spaces. Taking the cloud models and, when needed, pushing them into local environments.

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Problem

Stanley Black and Decker finds it unsustainable to ingest, transmit, store, query and analyze all data generated at the edge and more specifically on construction sites or rural areas with constrained network resources.

Solution

AWS Greengrass enables Stanley Black and Decker to monitor and filter data at the edge of the network enabling applications to send asset health and predict any mechanical failures before they occur. Edge-based applications built on AWS Greengrass will help us detect and compare vibrations emitted by high value tools to historical signatures that indicate everything from normal operations to imminent failure.

Impact

Instead of trying to use all the data Stanley Black and Decker will utilize AWS Greengrass allows us to focus on the right data. Some of the applications we see this fit includes remote troubleshooting of hydraulic assets by technicians, maintenance interval tracking, fuel savings, and alerts.

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Problem

As Konecranes specializes in the manufacturing and service of cranes globally, they discovered that when they needed to make updates to their machinery it meant downtime and local presence onsite.

Solution

AWS Greengrass has enabled them to deploy updates using cloud models that continually get smarter over time as they sync with the local environments.

Impact

This allows them to simplify their current crane architecture and make it possible to update calculations to the cranes in a secure way even after the installation has taken place.

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Who is AWS Greengrass for?

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Partner ecosystem

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AWS Greengrass launch partners

accenture
acer
Aricent
Ayla
BCG Digital Ventures
Bright Wolf
BSQUARE
CANONICAL
ctp Cloud Technology Partners
DIGI
enel
intel
lenovo
lightbits labs
LUXOFT
machineshop
MONGOOSE OS
NOKIA
PENTAIR
QNAP
QUALCOMM
RaspberryPi
resin.io
RioTinto
Saguna
SAMSUNG
solstice
StanleyBlack&Decker
ST life.augmented
technicolor
TELUS
ThingLogix
ThunderSoft
trek10
wistron
amazon
annapurnalabs an amazon company
AWS Snowball Edge

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AWS Greengrass Development

AWS Greengrass Core

Building the ecosystem

Routing messages

Working with device shadows

Local compute with Lambda

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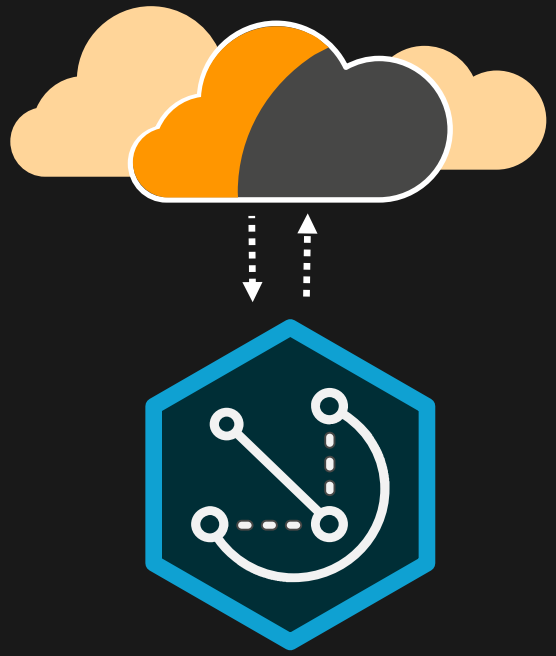
AWS Greengrass Core

Runs outside AWS – in physical environments
Composed as an operating environment
Consumes centralized configuration from AWS
TLS 1.2 mutual auth with AWS IoT
Integrates with virtually any AWS service
Interoperates with the local OS and ecosystem
Executes on demand and long running Lambda functions



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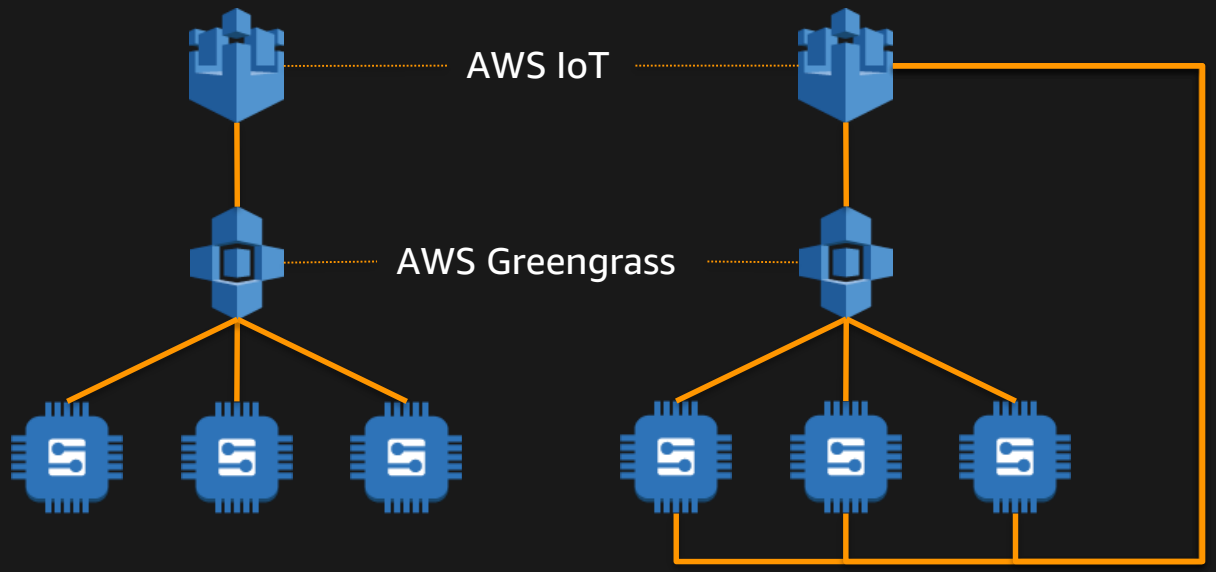
Building the ecosystem



No Auto-Discovery

Closed M2M networks

Auto-Discovery

Open M2M networks

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Routing Greengrass messages

Devices

Lambda Functions

AWS IoT

Device Shadows

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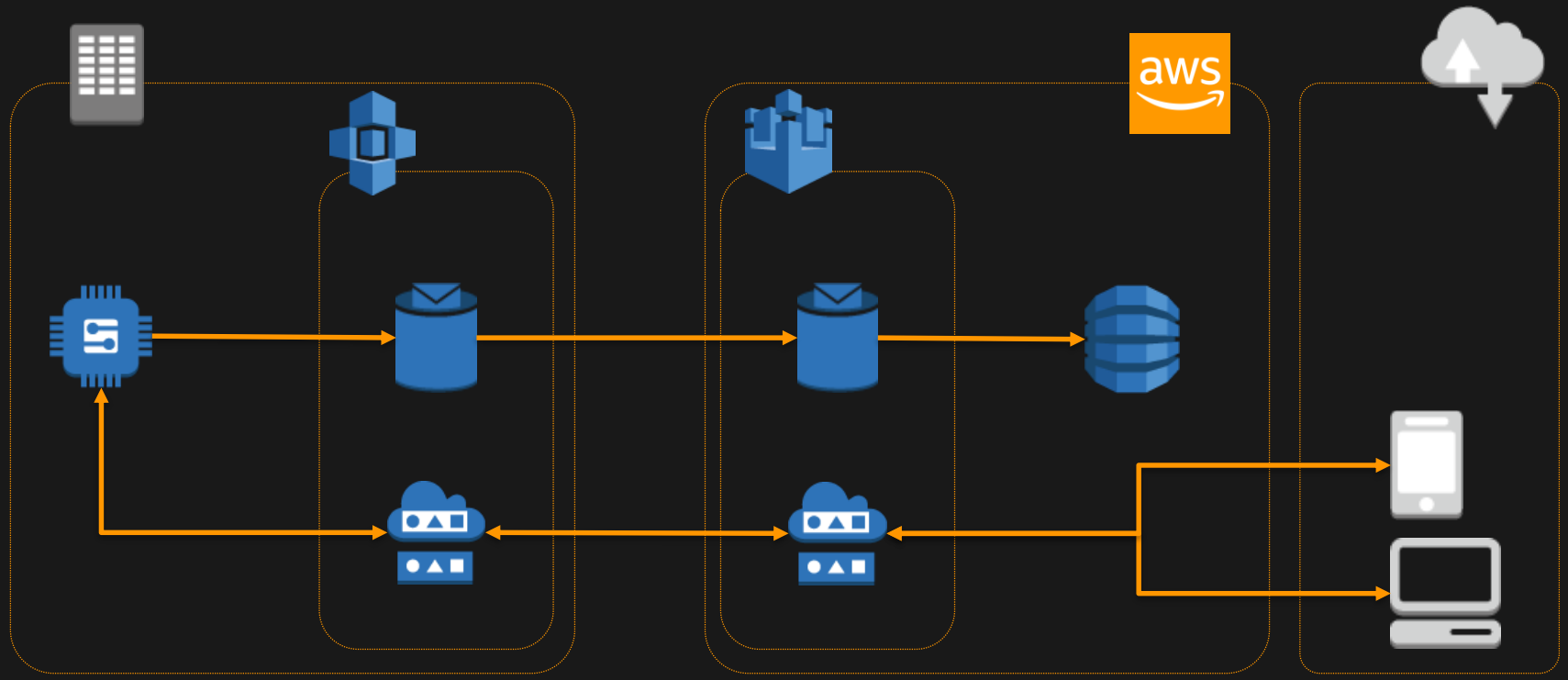
Device shadows with Greengrass



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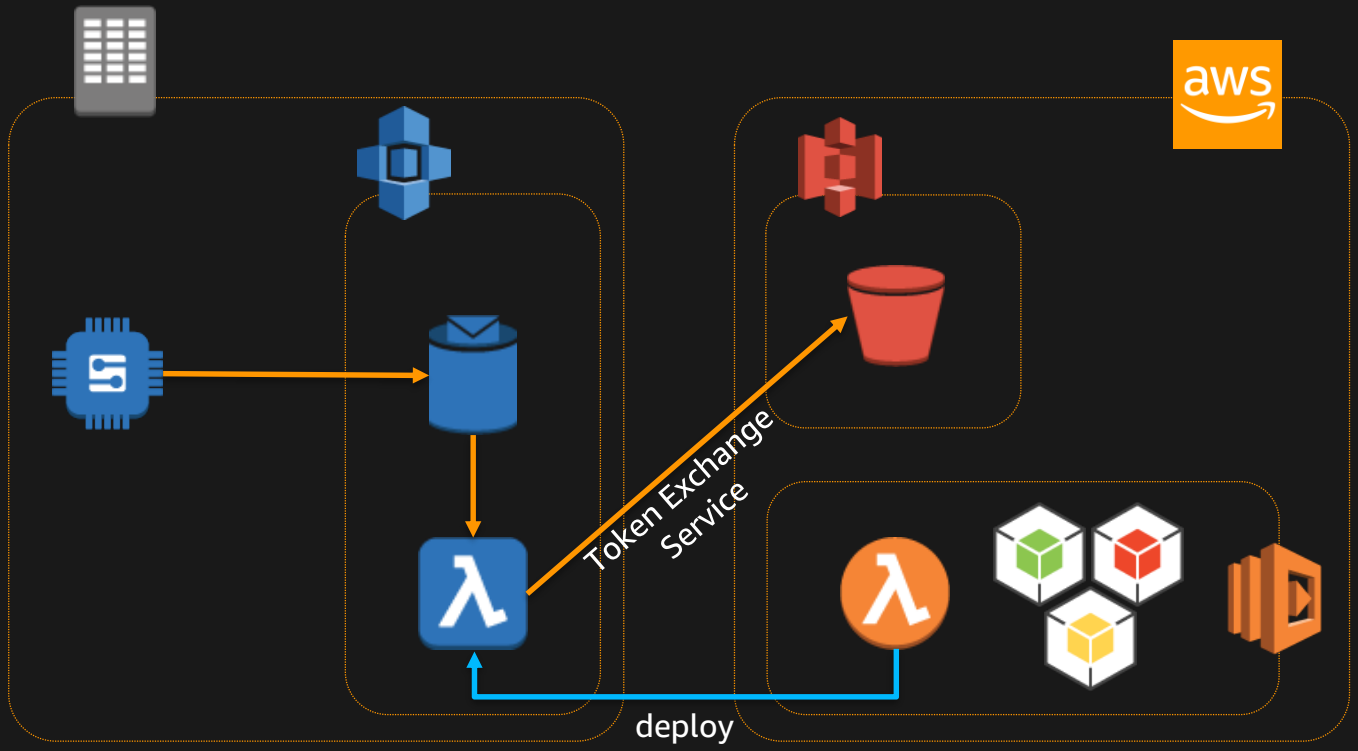
Using Lambdas in Greengrass



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AWS Greengrass at Scale

Hardware

Operating systems

Manufacturing and going to market

Ongoing maintenance and deploying Greengrass Groups

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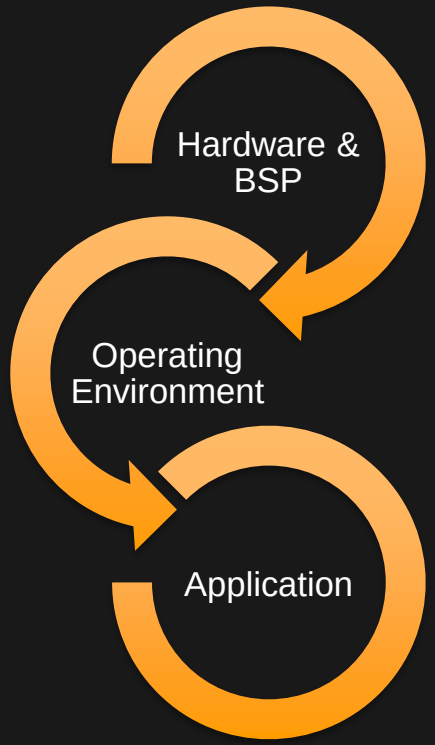
Three delivery lifecycles

hardware engineering

system engineering

software engineering



very rarely upgraded

rarely upgraded

Iteratively upgraded

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Key hardware and OS decision criteria

Specialized or commodity hardware

Specialized or commodity OS

Environmental, safety, and regulatory considerations

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What does specialized mean for hardware?

- Compute
- Connectivity
- Security
- Storage

- Reference
- Specialized
- Pluggable peripherals

- PCB
- Mechanics
- Enclosures

- Global supply chain
- Scale up – scale out
- Flashing

- Distribution channels
- Branding
- Integrator ready

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What does specialized mean for OS?

- Power
- Size
- Resiliency
- Skills
- Support

- Monolithic
- Modular
- Micro (RT)

- Upstream
- Downstream

- Manufacturing
- Upgrades

- Firmware driven updates
- Package manager driven updates

Automation is the key to longevity

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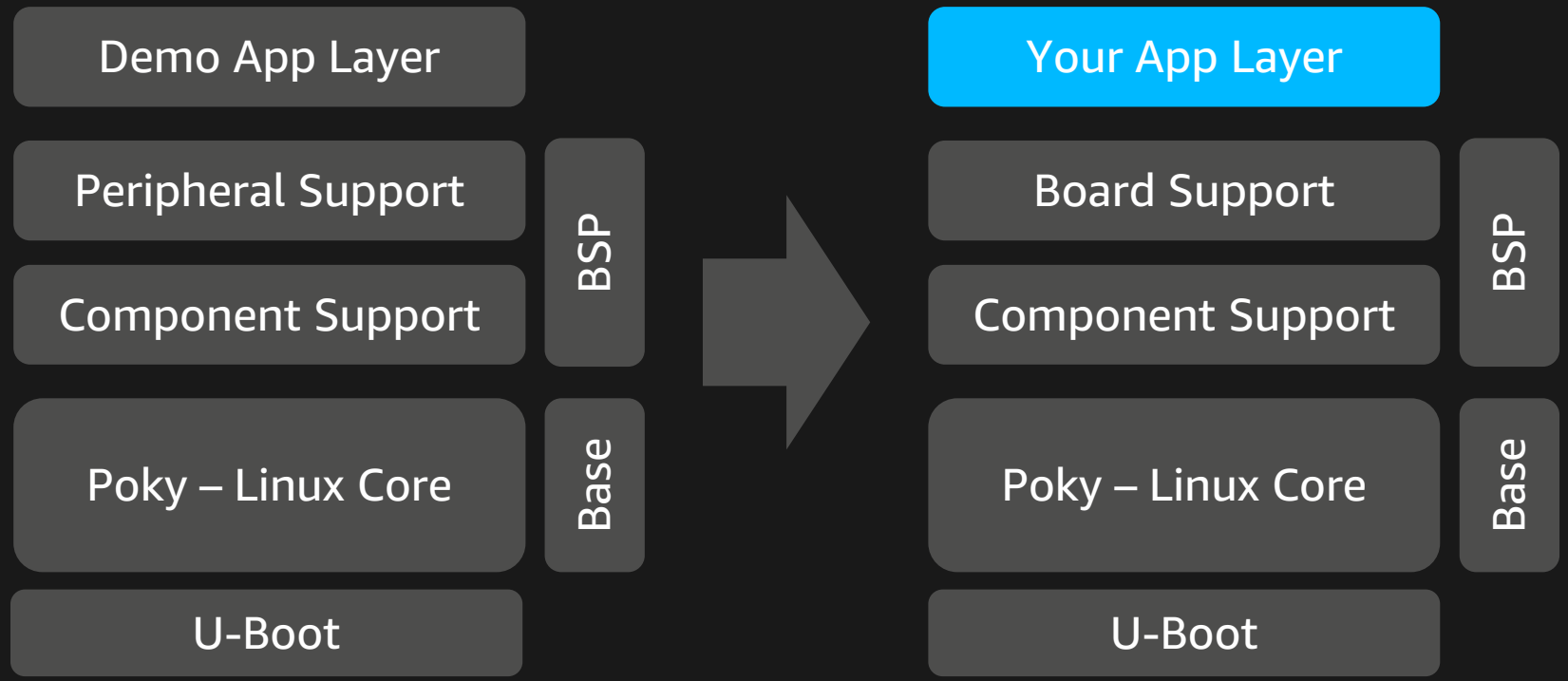
Yocto Embedded Linux – same board



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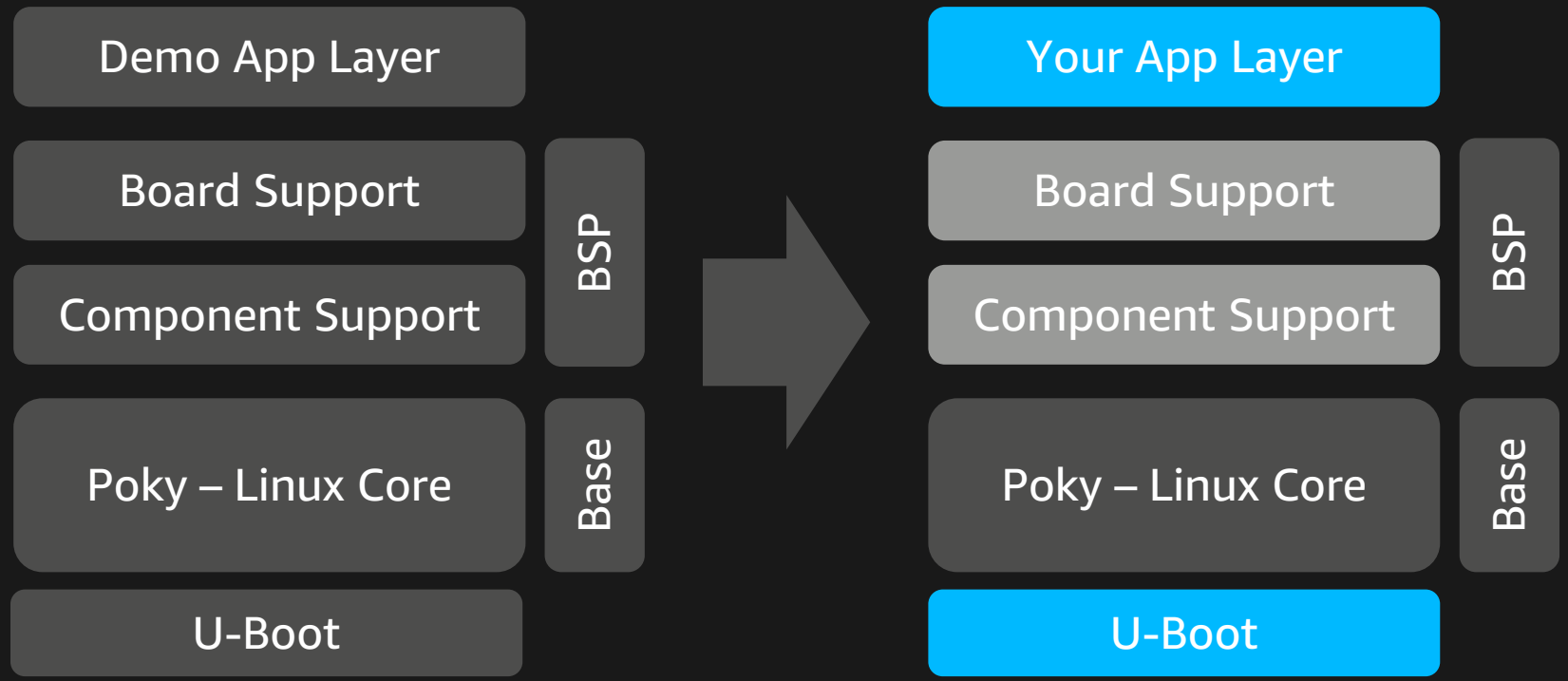
Yocto Embedded Linux – different board



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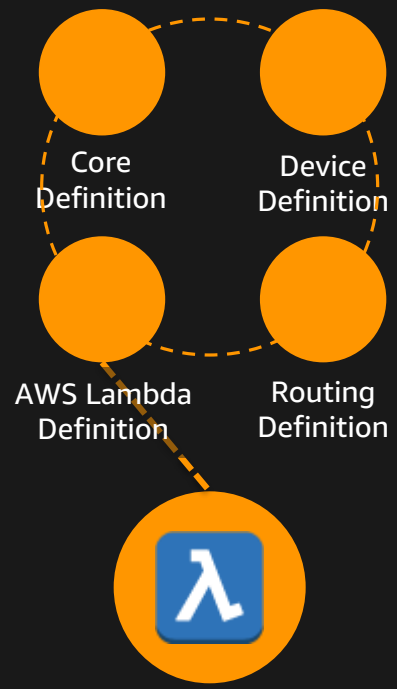
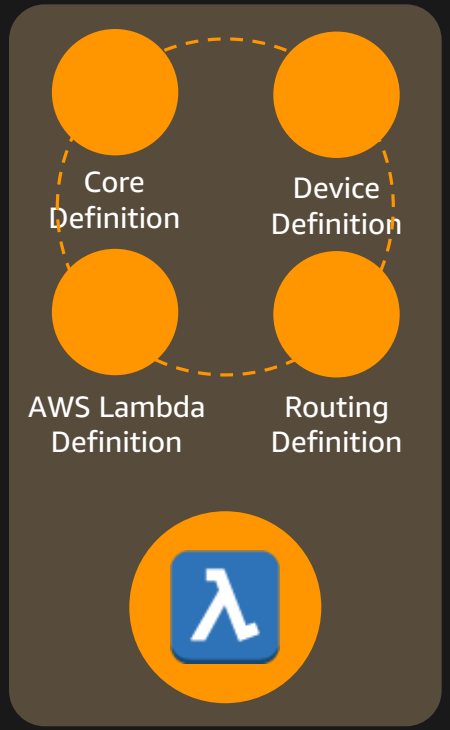
Greengrass group deployments





- Verifies group definition
- Verifies lambda function integrity
- Installs group.json and functions
- Restarts AWS Greengrass

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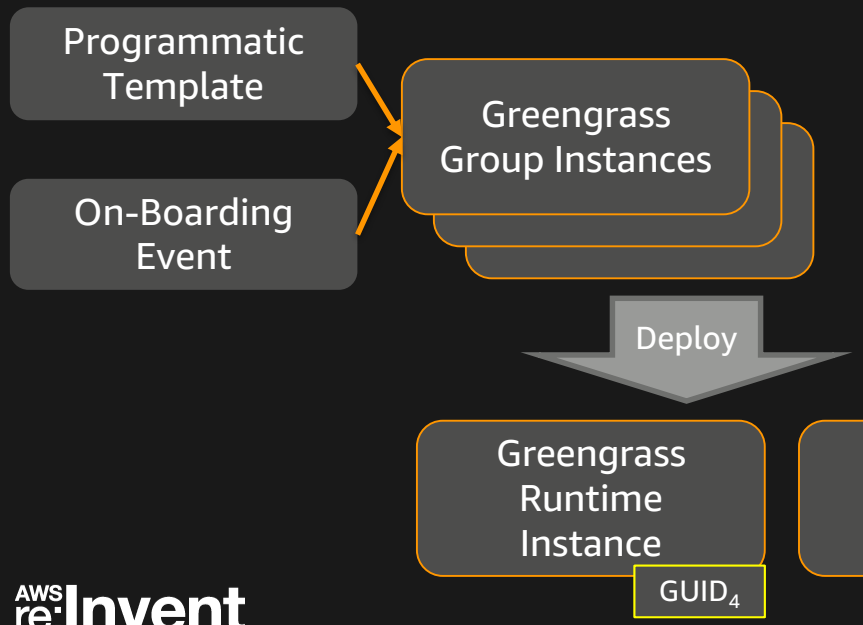
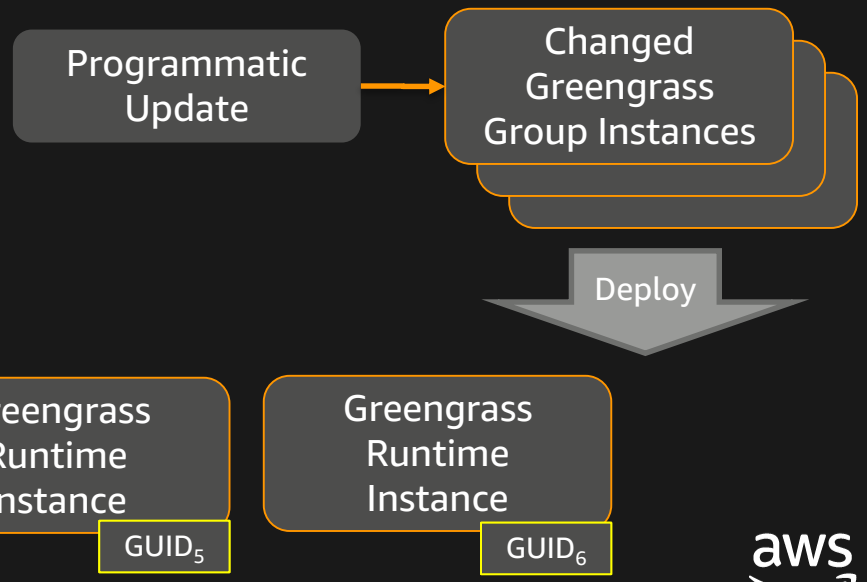
Scaling configuration deployments

New Greengrass Groups



Update Greengrass Groups



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Continuous Vital Signs Monitoring

Digi Healthcare Demo Built on AWS

Alex González

Supervisor, Software Engineering, Digi International

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Background

Scenario: Field healthcare as a result of a natural disaster

Challenges:

- Limited onsite medical staff
- Patient monitoring
- Mobile infrastructure

Technical solution:

- Smart patient node for each patient
- Smart hospital gateway for central monitoring
- Field to cloud connectivity using CAT1

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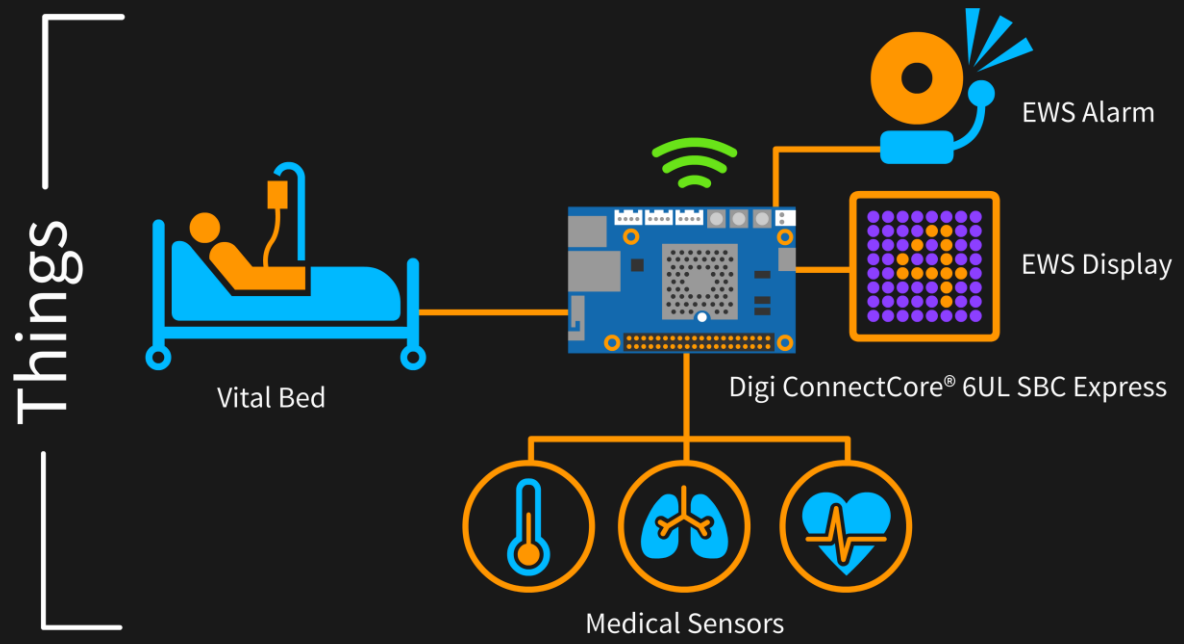
Smart Patient Node



The node composes all required commercial off-the-shelf components

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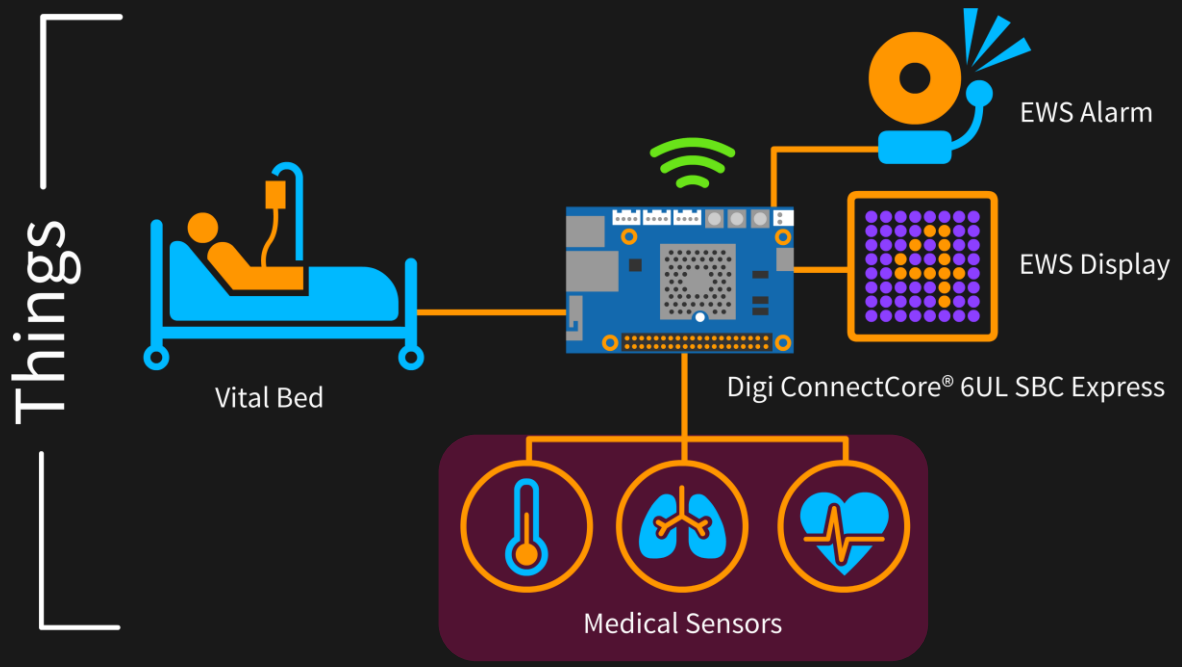
Smart Patient Node



Each patient has a bed device monitoring vital signs

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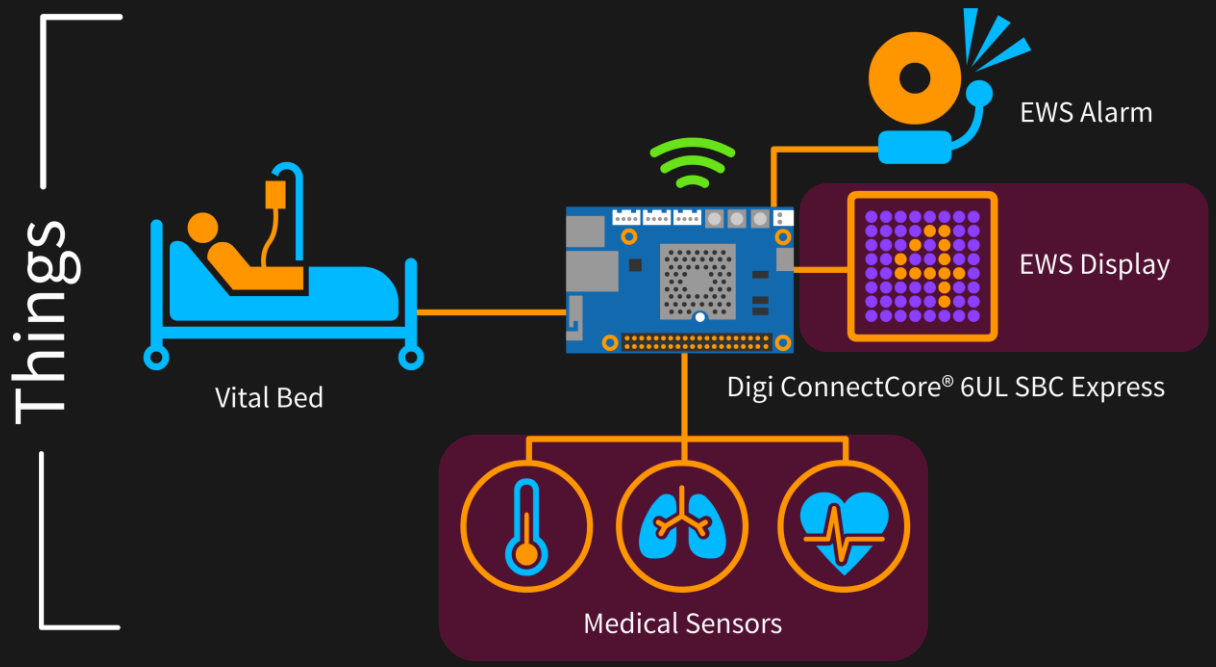
Smart Patient Node



The Early Warning Score (EWS) visually indicates the patient's condition

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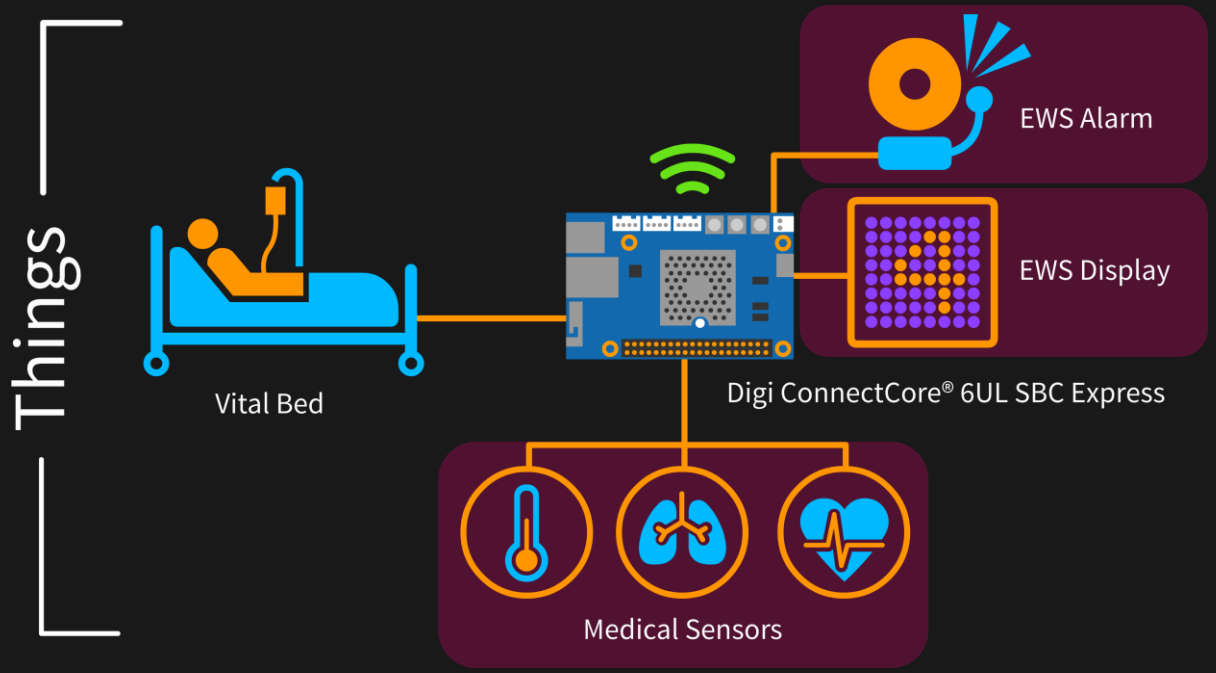
Smart Patient Node



The EWS alarm actuates noise locally and notifies staff

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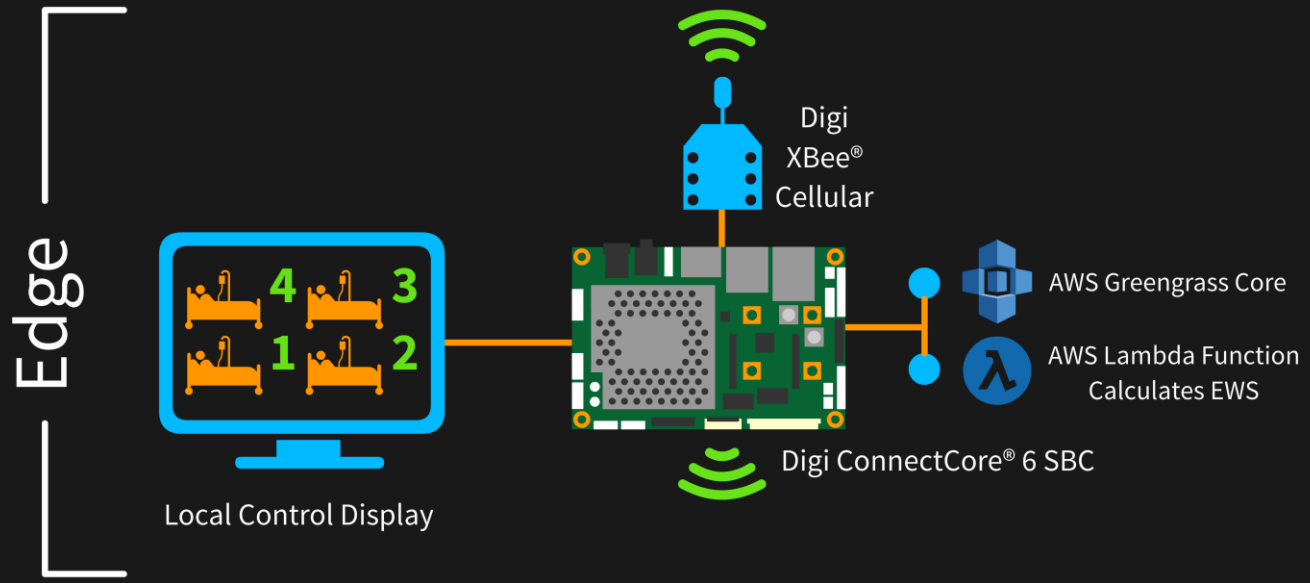
Smart Hospital Gateway



Intelligently routes and computes AWS Cloud and Greengrass Device data

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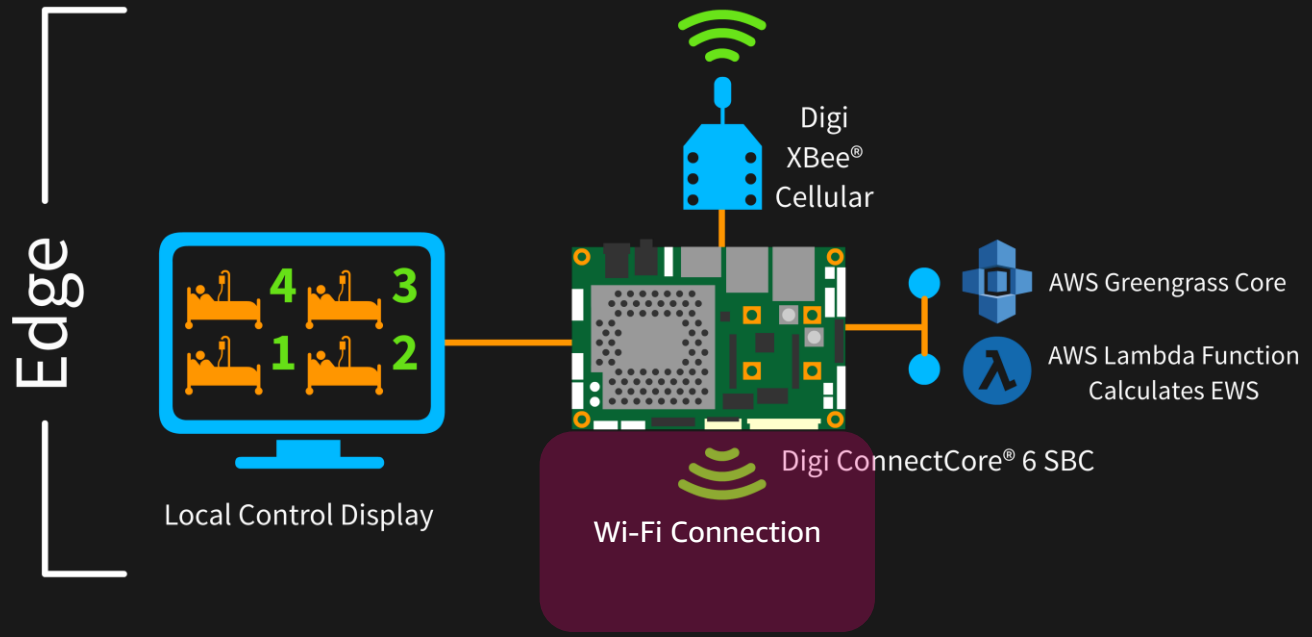
Smart Hospital Gateway



Securely receives data from bed devices

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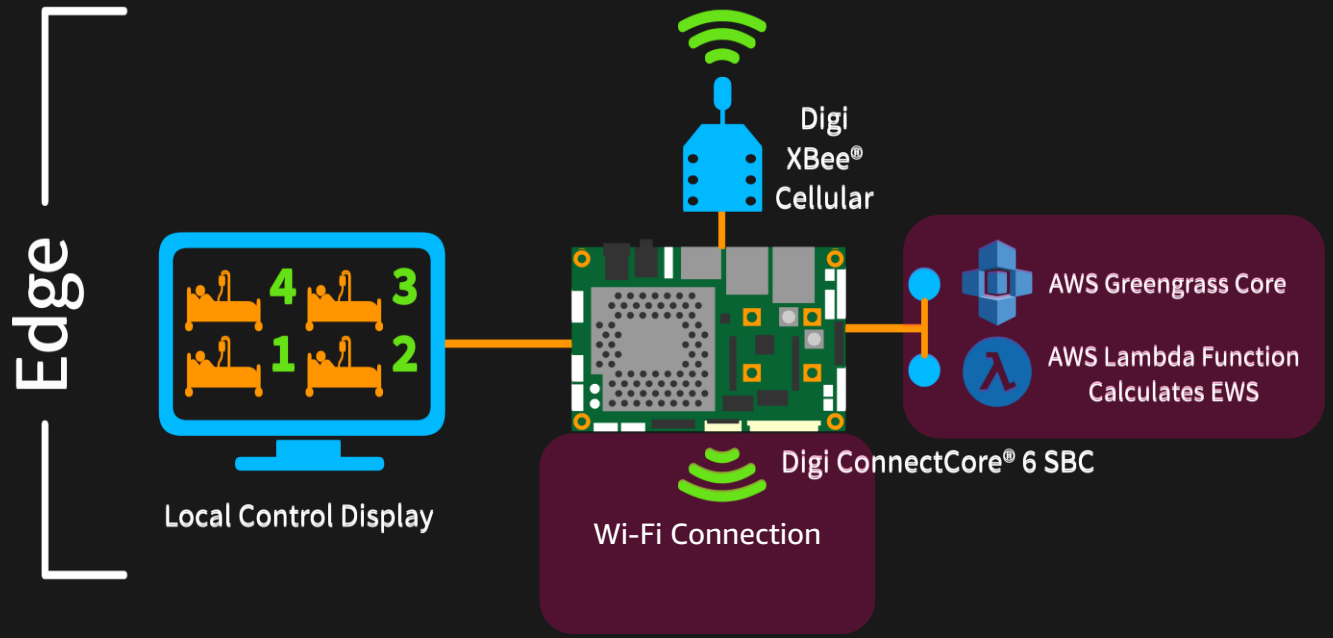
Smart Hospital Gateway



AWS Lambda computes the EWS based on patient profile and condition

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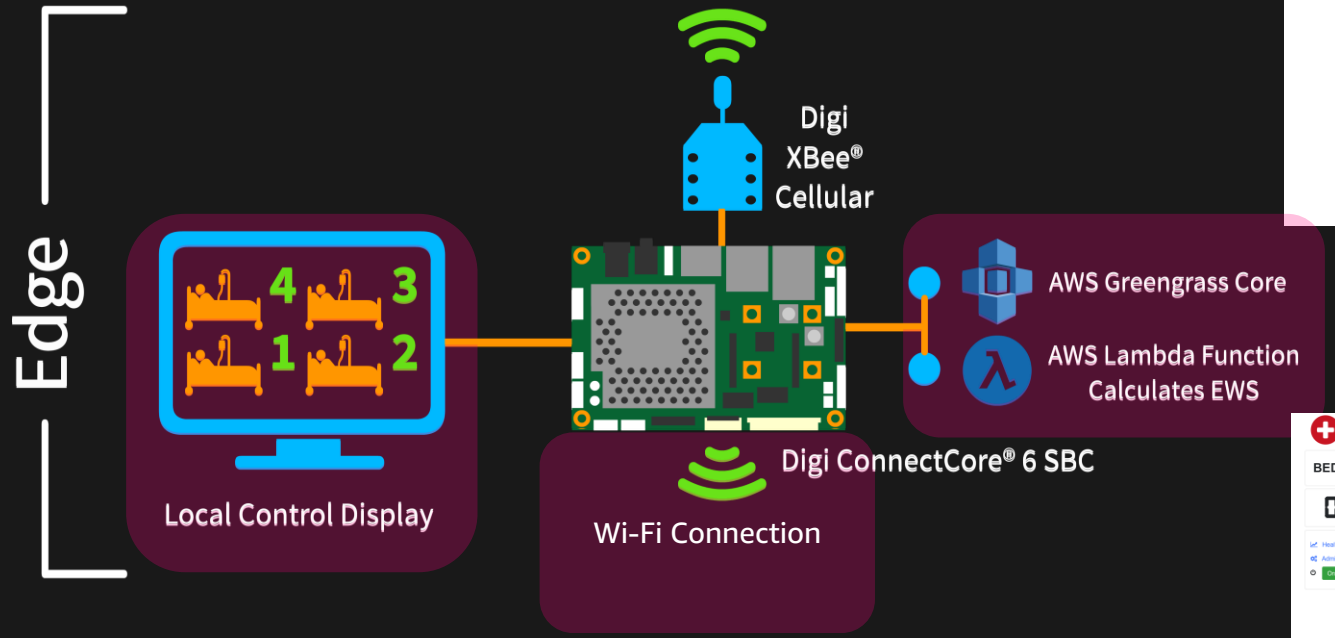
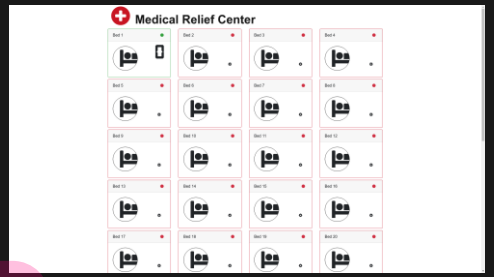
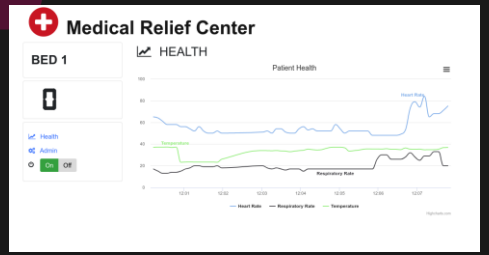
Smart Hospital Gateway







Displays critical data for central patient monitoring

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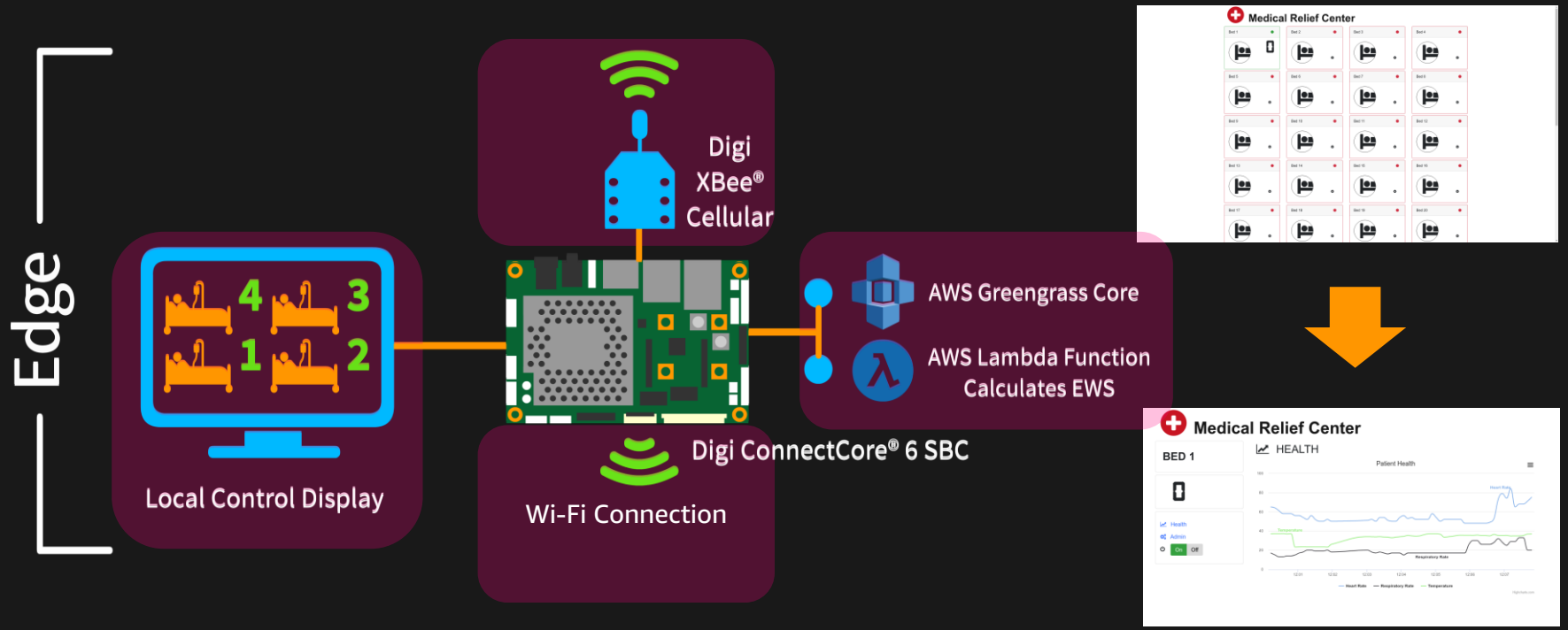
Smart Hospital Gateway



Securely transmits data to the AWS Cloud and can function offline

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AWS Cloud Analytics

AWS Greengrass securely sends patient data to AWS IoT

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AWS Cloud Analytics

AWS IoT consumes patient data and device state

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AWS Cloud Analytics

Safely stores data in Amazon DynamoDB for near real time Analytics

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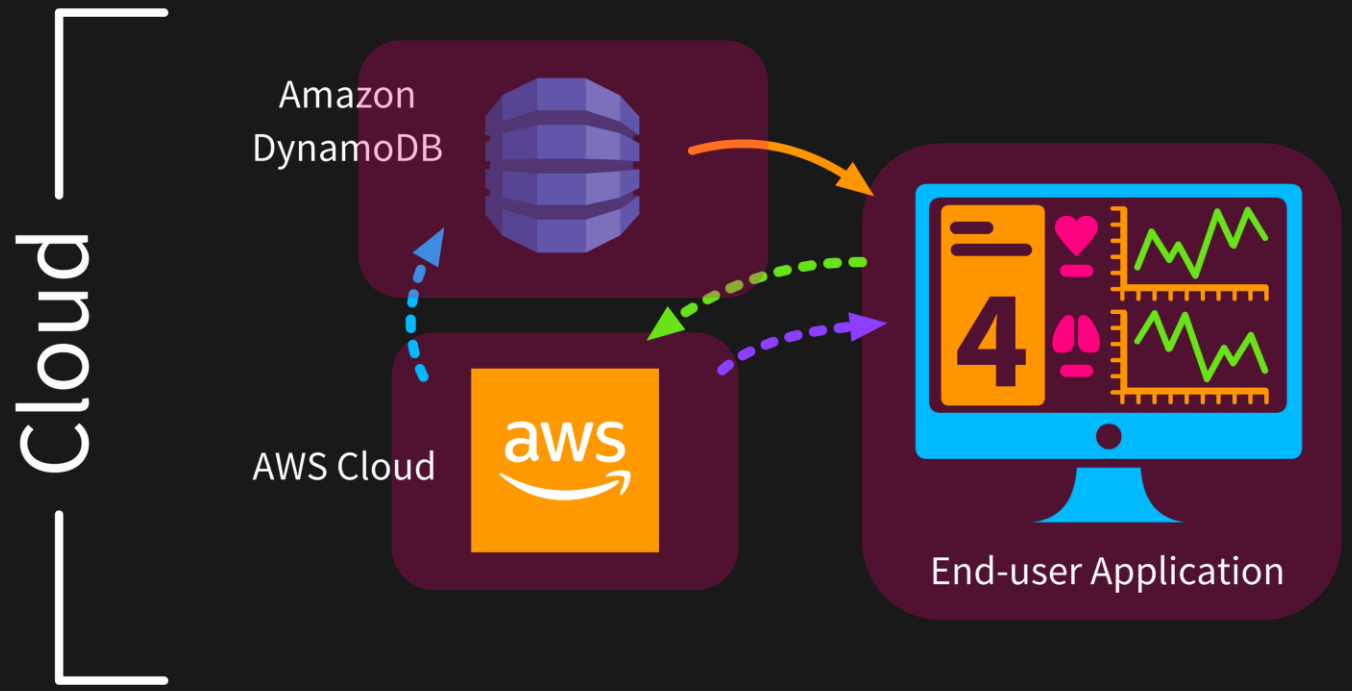
AWS Cloud Analytics



Amazon Cognito secures the web application's global reach

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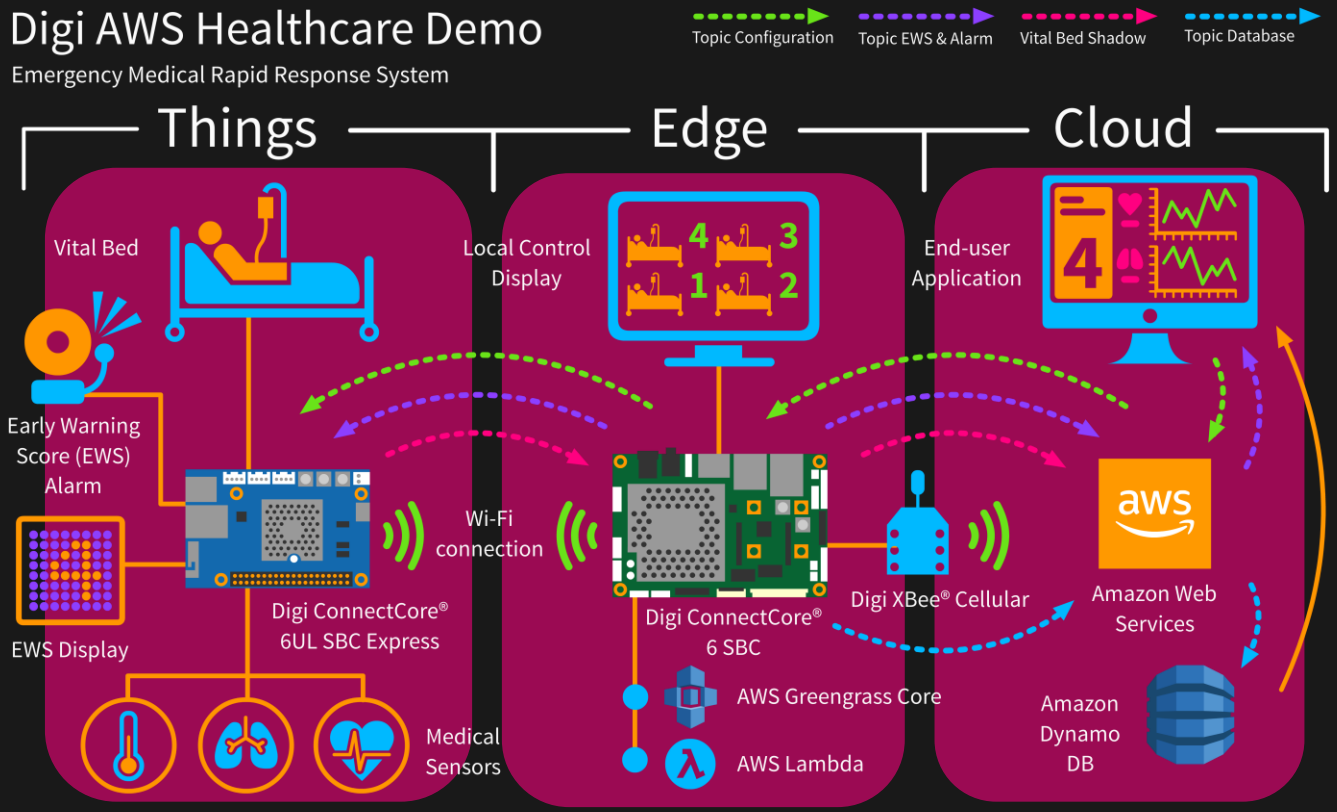
Digi AWS Healthcare Demo

Emergency Medical Rapid Response System



A ConnectCore 6UL SBC Express as the bed device to which, sensors, buzzer and small display are connected.

A ConnectCore 6 SBC as the edge device runs the AWS Greengrass core.

Global reach Web Application enables worldwide monitoring and analytics.

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Wrapping Up

Securely connect autonomous systems to AWS
Leverage existing skills for edge development
Evaluate the supply and value chain for go-to-market
Be prepared for mass scale through automation

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Contacts and Other Sessions

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Digi International @re:Invent

The Quad at Aria, Booth 209

AWS Greengrass Sessions

IOT309 - Compute at the Edge with AWS Greengrass

IOT318 - AWS IoT and AWS Greengrass - How Low You Can Go

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THANK YOU!

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